

BLOCKCHAIN FORENSIC ATTRIBUTION APPLIED TO THE ANALYSIS OF ADDRESSES ASSOCIATED WITH RUSSIAN INTELLIGENCE AGENCIES

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Belo Horizonte-MG, 20/08/2026

ABSTRACT

This study presents a reproducible blockchain forensic analysis of Bitcoin addresses associated with Russian intelligence agencies in the btc-battle dataset, combining on-chain evidence, graph metrics, and official national-security documentation. The core research question is whether public transaction traces can support operational inferences without overreaching into direct legal attribution of individuals, institutional command, or strict state responsibility. The adopted hypothesis is that the combination of technical observation (transactional data), quantitative derivation (centralities, flows, and temporality), and external attribution (institutional sources) enhances investigative prioritization and financial intelligence capabilities while preserving structural uncertainties. The methodology employs a Python notebook pipeline with data integrity controls, public API enrichment, confidence-graded evidence classification, directed graph modeling, and dual economic valuation (historical USD at transaction time and current-price USD). Findings consolidate 1,011 labeled addresses, 11,485 financial relations, and 855 nodes in the analyzed graph. Total identified volume reached 80,511,640,989 satoshis (805.1164 BTC), corresponding to USD 11,511,595.92 in historical transaction context and USD 56,066,696.55 in current price reference. Results indicate flow concentration in a few vertices, agency asymmetry, high-activity temporal windows, and operational relevance of intermediary nodes. It is concluded that blockchain forensics provides robust technical evidence for investigative triage and correlation, provided it is accompanied by epistemological caution, methodological traceability, and triangulation with external sources.

Keywords: Blockchain Forensics. Cyber Attribution. Russian Intelligence. Election Interference. Cryptoassets. Graph Analysis.

1. INTRODUCTION

The advancement of geopolitically motivated cyber operations has elevated the role of digital financial trails in the investigation of advanced persistent threats (APTs). In cases with potential state involvement, cryptocurrency flows can serve as indicators of logistical support, operational financing, infrastructure procurement, and resource movement among technical actors. This scenario demands forensic techniques that reconcile mathematical rigor, interpretive prudence, and transparent documentation.

In the case of the Russian Federation, government assessments and public indictments by the United States have highlighted coordinated influence and cyber intrusion activities in electoral processes, including direct references to the use of Bitcoin for acquiring technical infrastructure (UNITED STATES, 2017; UNITED STATES OF AMERICA, 2018; UNITED STATES, 2020). Concurrently, regulatory and sanctioning measures have increasingly emphasized the traceability of virtual assets as a risk mitigation instrument (UNITED STATES, 2020b).

This study is anchored in the **btc-battle** repository (*Bitcoin Battle: Burning Bitcoin for Geopolitical Fun and Profit*), which provides a dataset of 1,011 addresses labeled by agency and is tied to academic research accepted at IEEE ICBC 2025. Based on this input, an independent technical-forensic analysis was developed with reproducible execution and auditable artifact generation. The objective is not to 'prove criminal liability' self-sufficiently, but rather to structure robust quantitative evidence for investigative prioritization and analytical threat support.

1.1 Research Problem

To what extent does blockchain forensic analysis, applied to addresses labeled as associated with GRU, FSB, and SVR, allow the inference of relevant operational patterns without overstepping the boundaries of evidentiary validity?

1.2 Hypothesis

The central hypothesis is that the integration of graph metrics, temporal analysis, and financial valuation generates consistent signals for threat intelligence, provided that each conclusion is classified by level of evidence and accompanied by explicit uncertainty boundaries.

1.3 Objectives

General Objective: Apply a reproducible blockchain forensics protocol to examine financial relationships involving addresses associated with Russian intelligence agencies.

Specific Objectives:

1. Consolidate and validate the address and transaction database through public enrichment;
2. Construct a directed graph and extract structural centralities;
3. Quantify flows across temporal windows and by agency;
4. Compare values across satoshis, BTC, historical USD, and current USD;

5. Differentiate observed, derived, and inferred evidence;
6. Discuss limitations, false-positive risks, and operational implications.

1.4 Scientific and Applied Relevance

Academically, this research contributes to the cyber conflict attribution debate under a probabilistic, multi-layered perspective. Practically, it delivers an auditable analytical workflow with reproducible artifact export and standardized terminology suitable for interoperability among technical teams, financial intelligence units, incident responders, and legal support.

2. DEVELOPMENT

2.1 Cybersecurity Attribution as an Uncertainty Problem

Specialized literature characterizes attribution as an incremental process subject to noise, plausible deniability, operational proxies, and observability constraints. Rid and Buchanan (2018) emphasize that attribution requires combining technical signals, strategic context, and probabilistic inference, avoiding deterministic conclusions drawn from incomplete data.

2.2 Official Evidence on Russian Activities and Electoral Environment

The ODNI report (2017) outlines Russian influence campaigns in the U.S. electoral environment. The DOJ indictment (2018) against GRU officers details the use of Bitcoin for acquiring digital infrastructure. The U.S. Senate Intelligence Committee report (2020) consolidates institutional vulnerabilities and exploitation patterns. These documents contextualize working hypotheses and calibrate forensic interpretation.

2.3 Regulatory Dimension and Financial Intelligence

Enforcement actions by the U.S. Department of the Treasury (OFAC) reinforce that virtual assets fall directly under the scope of monitoring illicit financing and sanctions evasion. Methodologically, this justifies incorporating flow analysis, asset concentration, and financial intermediation as core risk-screening components (UNITED STATES, 2020b).

2.4 Primary Technical Source

The **btc-battle** repository explicitly documents the scope of the address dataset and its linkage to research accepted at IEEE ICBC 2025. This thesis utilizes

this dataset as a foundational baseline and expands observability through public transactional enrichment, maintaining strict separation between: (a) origin labeled data; (b) proprietary computational derivations; and (c) analytical interpretations.

2.5 Methodology

Research Type: Applied, quantitative-computational research with documentary triangulation and forensic orientation. The study avoids invasive techniques or private data collection, operating entirely on public ledgers and reproducible artifacts.

Technical Environment and Reproducibility: The pipeline was implemented in the notebook ``bitcoinbattleforensicanalysis.ipynb``, with dependencies specified in ``requirements-notebook.txt``. Structured tabular outputs are persisted in CSV format within the ``results/`` directory, and graphical artifacts are consolidated in ``thesisfigures/``.

Analytical	Project	Repository:
https://github.com/dezugin/russianintelligencebtc_analysys		

Pipeline Stages:

- 1. **Ingestion and Integrity:** Ingestion of labeled address datasets and baseline validation;
- 2. **Transactional Enrichment:** Collection of on-chain transactions via public APIs with temporal normalization;
- 3. **Graph Modeling:** Construction of weighted directed networks (source → destination);
- 4. **Structural Metrics:** Calculation of PageRank, betweenness centrality, and connectivity metrics;
- 5. **Temporal Analysis:** Monthly and daily aggregations of satoshis and USD (historical/current);
- 6. **Evidence Classification:** Categorization into observed, derived, inferred, and external attribution;
- 7. **Forensic Export:** Generation of auditable tables and vector charts for forensic tracking.

Evidence Classification Model: To mitigate confirmation bias and overinterpretation, findings were formally categorized into four certainty classes: **OBSERVED** (directly verifiable on-chain or in raw input datasets); **DERIVED** (mathematical computations over observed data); **INFERRED** (operational

hypotheses derived from behavioral patterns); and **EXTERNAL_ATTRIBUTION** (contextual data from institutional reporting). Strong conclusions are strictly limited to instances of multi-source convergence.

A Priori Methodological Boundaries:

- 1. An address does not inherently equate to an individual or legal entity;
- 2. Wallet reuse, mixing protocols, CoinJoin, and custodial services can distort graph topology;
- 3. Lack of off-chain metadata prevents deterministic confirmation of command-and-control hierarchies;
- 4. Absence of heuristic expansion reflects deliberate conservative parameter tuning to avoid false positives.

2.6 Dataset Characterization and Baseline Scope

The consolidated quantitative scope encompasses:

- Total labeled addresses: **1,011**;
- Agency distribution: **GRU (1,004), FSB (4), SVR (3)**;
- Enriched transaction edges: **11,485**;
- Distinct nodes in resulting graph: **855**.

The heavy concentration of labels in the GRU category reflects differential intelligence visibility and the collection parameters of primary sources, rather than an absolute measure of comparative operational intensity between agencies.

2.7 Financial Flow Analysis and Economic Valuation

The aggregated financial volume identified in the analyzed transaction network reached:

- **80,511,640,989 satoshis**;
- **805.1164 BTC**;
- **USD 11,511,595.92** (historical valuation at transaction execution time);
- **USD 56,066,696.55** (converted at current reference pricing).

The dual valuation is essential in financial forensics: historical USD reflects economic purchasing power and operational capability at the time of execution, whereas current USD measures contemporary asset materiality.

Table 1 – Top 5 months by historical USD volume.

Month	Historical USD (\$)	Current USD	Satoshis
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Equivalent (\$)			
2022-02	2,863,006	4,818,739	6,919,697,355
2021-10	1,589,031	1,924,978	2,764,263,955
2020-09	974,769	6,625,587	9,514,326,632
2021-11	854,172	888,204	1,275,458,767
2020-02	753,162	5,706,872	8,195,054,875

Source: Research data (2026).

2.8 Network Structural Results and Centrality Metrics

Centrality metrics indicate that a small fraction of nodes accumulates disproportionate structural importance within the network. This property is characteristic of financial networks with centralized intermediation, where monitoring key hubs significantly impacts analytical coverage.

Table 2 – Highest PageRank values (Top 10 sample).

Address	PageRank	Total Degree	Counterparts	Historical USD (\$)	Current USD (\$)
1JoHq7tYLuBZw NNh1VUyoevejF zVsu3WRG	0.027675	286	286	360.86	715.69
1DLA46sXYps3 PdS3HpGfdt9Mb Qpo6FytPm	0.024663	288	288	430.36	1,228.85
1L5QKvh2Fc86j 947rZt12rX1EFr CGb2uPf	0.017575	289	289	4,556.36	28,422.40
16chcAK75pcNq oCd9e1EZ6XkCt yVnwTfpD	0.016202	278	278	4,634.05	53,271.60
39eFkkjKpH1jV7 ELzU7Hk7TSerN dXwVyTU	0.016202	278	278	490.31	1,860.67
3FtaEu1orGLafs 47gwrp8RyWyvt pLPL4n9	0.016063	278	278	1,055.56	5,594.02
3QRgXDZT3xyM m27iGnZeYHnn GyoSBRX7d1	0.016029	279	279	555.87	2,786.62
1EBSUw4VdEth wSTeGrEQCbfs EMJXpwwAVp	0.010949	1	1	20.48	188.98
1Pf3fCAkmFH5T rNngmKs9th9f75 xNkHovQ	0.008739	95	94	103,574.71	177,197.01
3DXBaSW2N5P XJTwsGeuqM44 NQEWLNMYoco	0.008521	7	7	27,277.79	332,760.62

Source: Research data (2026).

Table 3 – Highest Betweenness Centrality values (Top 10 sample).

Address	Betweenness	Total Degree	Counterparts	Historical USD (\$)	Current USD (\$)
1cpu7jjzht9FQ Y8aRz26Uc6j8c prW7Zz	0.000465	70	69	47,453.07	315,671.07
1G6kVJsdK5NS FvVa3sKvuAzj92 EK4yq1nZ	0.000292	39	38	68,163.05	219,331.51
3QRgXDZT3xyM m27iGnZeYHnn GyoSBRX7d1	0.000290	279	279	555.87	2,786.62
16chcAK75pcNq oCd9e1EZ6XkCt yVnwTfpD	0.000289	278	278	4,634.05	53,271.60
39eFkkjKpH1jV7 ELzU7Hk7TsErN dXwVyTU	0.000289	278	278	490.31	1,860.67
3FtaEu1orGLafs 47gwrp8RyWyvt pLPL4n9	0.000289	278	278	1,055.56	5,594.02
1PzhMfvEdU5Ar h2zZfzUVHKLUC 2GsMuUws	0.000186	36	35	66,221.23	213,166.63
1L5QKvh2Fc86j 947rZt12rX1EFr CGb2uPf	0.000151	289	289	4,556.36	28,422.40
1DLA46sXYps3 PdS3HpGfdt9Mb Qpo6FytPm	0.000150	288	288	430.36	1,228.85
1Pf3fCAkmFH5T rNngmKs9th9f75 xNkHovQ	0.000145	95	94	103,574.71	177,197.01

Source: Research data (2026).

Operational Interpretation of Centrality Metrics: High PageRank indicates addresses frequently referenced across the global transaction graph. High Betweenness points to bridging infrastructure between clusters. For cyber intelligence operations, these nodes represent prime candidates for priority tracking, infrastructure correlation, and multi-source enrichment.

2.9 Attribution Expansion Strategies and Methodological Discussion

In the evaluated dataset, automated clustering and expansion heuristics (common input ownership, 1-hop, 2-hop) resulted in zero unauthorized growth:

Table 4 – Results of attribution expansion strategies.

Expansion Strategy	Seeds	Expanded	New Nodes	Growth Rate
seed_only	0	0	0	0.0%

common_input_ownership	0	0	0	0.0%
one_hop_transactional	0	0	0	0.0%
two_hop_transactional	0	0	0	0.0%
published_cluster_membership	0	0	0	0.0%

Source: Research data (2026).

Methodological Interpretation: This result validates the strictness of the forensic pipeline, which deliberately avoided automated scope inflation without corroborating off-chain evidence, minimizing false-positive associations.

High-Confidence Findings: (1) Non-trivial financial network associated with target entities; (2) High structural concentration in few key vertices; (3) Distinct temporal peaks of activity; (4) Substantial economic variance based on valuation timestamp.

Limitations on Direct Conclusion: (1) Civil identity of key controllers; (2) Formal state command chain per transaction; (3) Precise operational purpose without auxiliary intelligence.

2.10 Advanced Intelligence Analyses (Ransomware, OP_RETURN, OSINT)

This section consolidates advanced forensic modules: ransomware/Conti correlation, OP_RETURN metadata recovery, OSINT tagpack triangulation, and mixer pivoting.

2.10.1 Ransomware / Conti Correlation Diagnostics

Cross-referencing state-attributed addresses against ransomware intelligence feeds (including the Conti group) revealed no direct operational overlap in the observed subgraph.

Table 5 – Coverage diagnostics for Ransomware/Conti correlation.

Diagnostic Metric	Value
Candidate state-attributed addresses	1,011
State addresses present in graph	313
Candidate ransomware/Conti addresses	7,606
Ransomware/Conti addresses in graph	0
State addresses with observed edges	313
Ransomware addresses with observed edges	0
State → Ransomware paths discovered	0

Source: results/paper_table_ransomware_coverage.csv

2.10.2 OP_RETURN Metadata Extraction

Scanning transaction outputs with local cache fallbacks successfully recovered 4,846 OP_RETURN messages across 222 blocks, proving that non-financial communication coexists alongside fund movements.

Table 6 – OP_RETURN scan summary.

Evidence Source	OP_RETURN Records	Scanned Files
Blockstream cache scan	4,846	222

Source: results/paper_table_op_return_scan_summary.csv

2.10.3 Public Attribution and OSINT Tagpack Triangulation

Integrating public tagpacks and OSINT sources classified 313 nodes with strong public attribution, while maintaining a conservative contextual confidence rating.

Table 7 – Public attribution and mixer exposure summary.

Metric	Value	% of Nodes
Total nodes in enriched context	855	100.00%
Strong public attribution	313	36.61%
Explicit external attribution	0	0.00%
Direct mixer exposure	0	0.00%
1-hop mixer exposure	0	0.00%
Expanded mixer exposure	0	0.00%
Low contextual confidence rating	855	100.00%

Source: results/paper_table_201_attribution_summary.csv

2.11 Temporal Dynamics and Transaction Semantics

This section evaluates two complementary dimensions: temporal cadence (first/last seen, active lifespan, bursts, campaign phases) and behavioral edge semantics (funding, peeling chain, consolidation, fan-out, fan-in, pass-through).

2.11.1 Temporal Lifespan, Cadence, and Campaign Phases

Temporal metrics demonstrate that the network operates primarily in passive baseline conditions interspersed with short, high-intensity operational bursts (peak campaign, capturing 91.94% of transactions across 3.34% of active days):

Table 8 – Summary of temporal address enrichment.

Temporal Metric	Value
Addresses with temporal enrichment	855
Median active lifespan (days)	0.0
90th percentile active lifespan (days)	0.1421
Median inter-transaction interval (hours)	0.0
Addresses with ≥ 1 burst window	296
Addresses without burst window	559
Maximum observed burst (transactions/window)	211

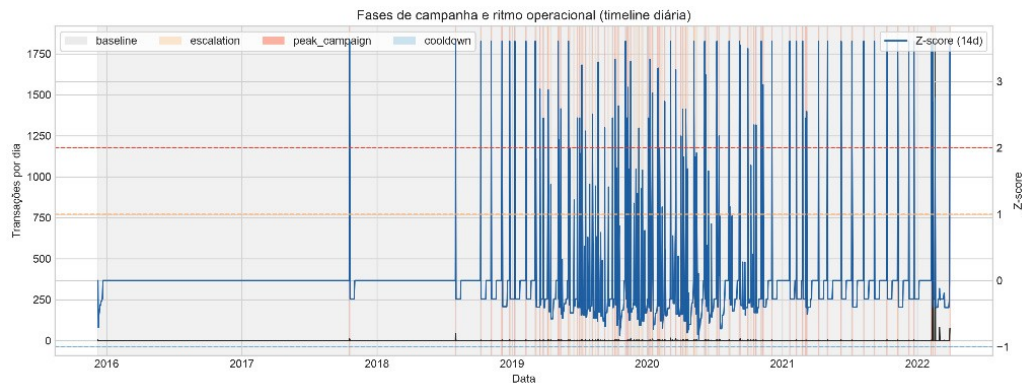
Source: results/paper_table_temporal_summary.csv

Table 9 – Campaign phase synthesis across daily series.

Campaign Phase	Windows	Total Days	Total Transactions	% of Days	% of Transactions
baseline	101	2,188	185	94.92%	4.52%
escalation	38	40	145	1.74%	3.54%
peak_campaign	75	77	3,765	3.34%	91.94%

Source: results/paper_table_campaign_phase_summary.csv

Figure 15 – Timeline of campaign phases and operational tempo.



Source: Authors (2026).

2.11.2 Transaction Semantics: From Edge Flows to Behavioral Typologies

Heuristic classification reveals that fan-out patterns dominate transaction frequency (89.11%), whereas initial funding transactions concentrate the vast majority of financial volume in BTC (610.94 BTC):

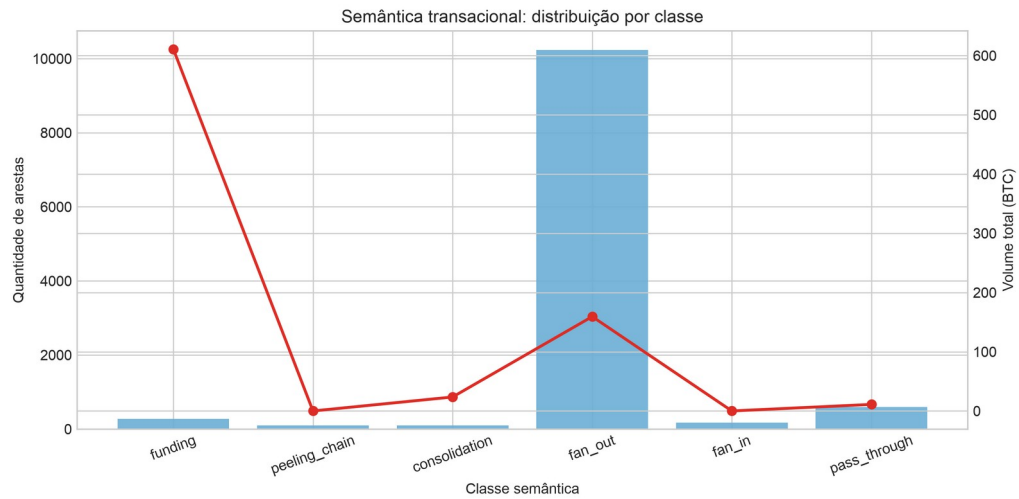
Table 10 – Distribution of transaction semantic classes.

Semantic Class	Edges	% of Edges	Total Volume (BTC)	Median per Edge (sats)
fan_out	10,234	89.11%	159.36193669	687.0

pass_through	596	5.19%	11.09491021	635,317.0
funding	278	2.42%	610.93952129	2,009,550.0
fan_in	178	1.55%	0.06204970	4,920.0
consolidation	100	0.87%	23.63355074	564,120.5
peeling_chain	99	0.86%	0.02444126	580.0

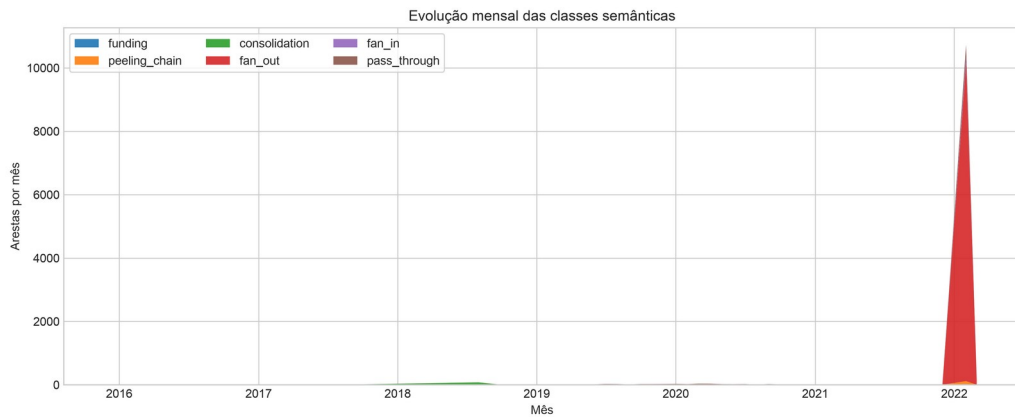
Source: results/paper_table_semantic_class_summary.csv

Figure 16 – Transaction semantic classes overview.



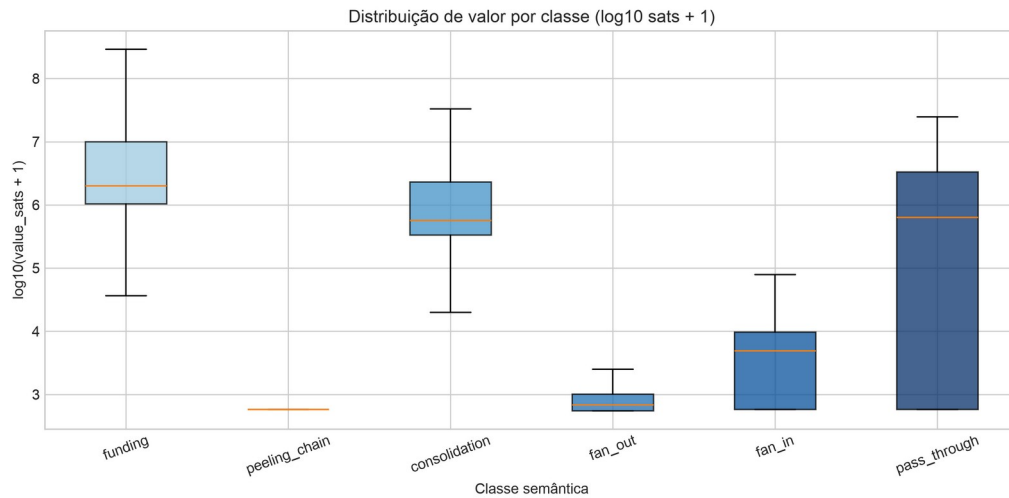
Source: Authors (2026).

Figure 17 – Monthly evolution of semantic edge classes.



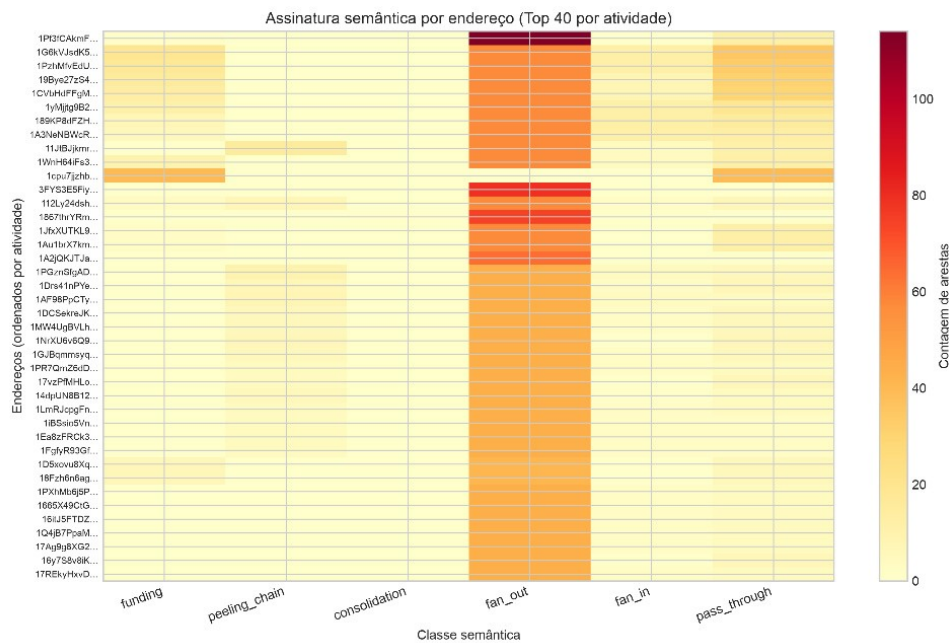
Source: Authors (2026).

Figure 18 – Financial value distribution by semantic class (log10 sats + 1).



Source: Authors (2026).

Figure 19 – Semantic signature heatmap per address (Top 40 active).



Source: Authors (2026).

2.12 Longitudinal Community Detection (Louvain, Leiden, and Infomap)

To transition from isolated edge tracing to recurring operational cells, community detection was applied across quarterly network snapshots using three complementary algorithms:

Louvain: A greedy modularity maximization algorithm optimizing partition density Q (BLONDEL et al., 2008):

$$Q_c = \frac{1}{2m} \sum_i \sum_j A_{ij} \mathbf{1}\{c_i = c_j = c\} - \left(\sum_i \frac{k_i}{2m} \mathbf{1}\{c_i = c\} \right)^2$$

$$= \frac{\Sigma_{in}}{2m} - \left(\frac{\Sigma_{tot}}{2m} \right)^2$$

Leiden: Introduces an explicit community refinement phase prior to aggregation, guaranteeing that discovered communities are internally well-connected and eliminating disconnected partition artifacts (TRAAG; WALTMAN; VAN ECK, 2019).

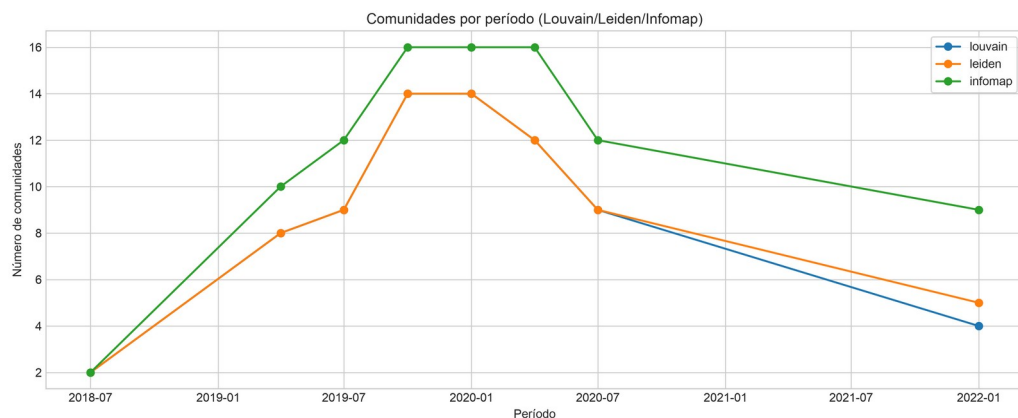
Infomap: Minimizes the description length of random walk trajectories (Map Equation), providing superior sensitivity to flow routing and passing infrastructure (ROSVALL; BERGSTROM, 2008).

Table 11 – Synthetic comparison of Louvain, Leiden, and Infomap over time.

Algorithm	Communities per Period	Largest Community	Modularity Range Q
Louvain	2 to 14	303 nodes	0.0764 to 0.8010
Leiden	2 to 14	299 nodes	0.0764 to 0.8010
Infomap	2 to 16	285 nodes	0.0748 to 0.7995

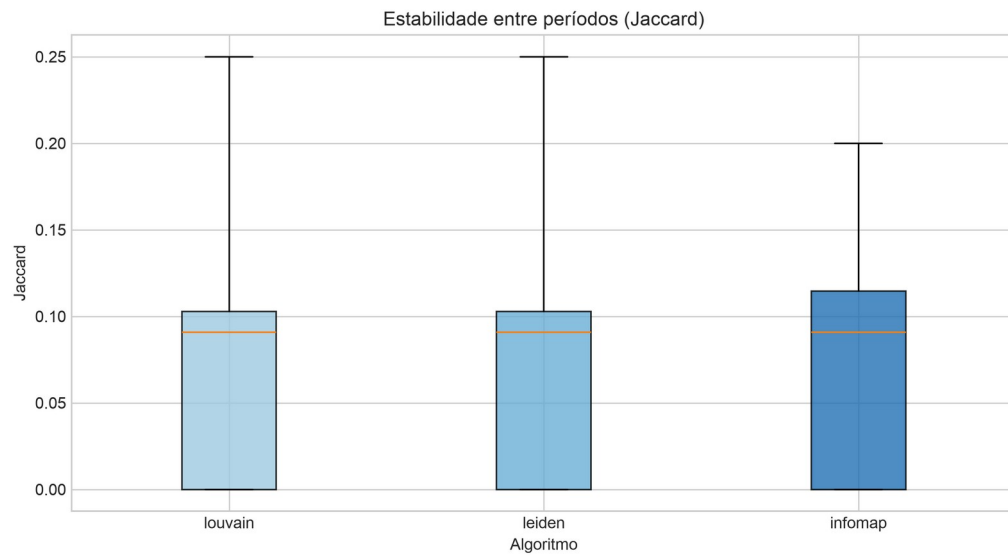
Source: results/community_period_summary.csv

Figure 20 – Community count over time across algorithms.



Source: Authors (2026).

Figure 21 – Temporal stability between periods (Jaccard Index).



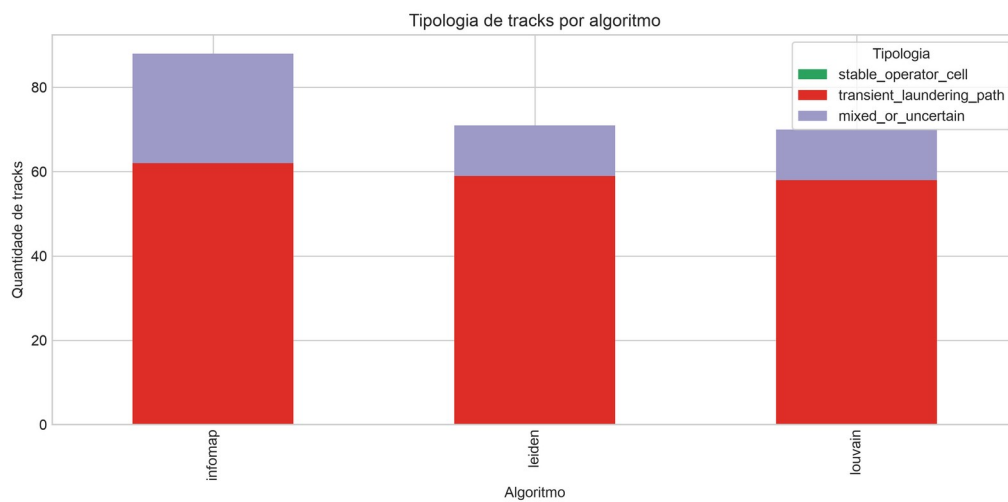
Source: Authors (2026).

Table 12 – Community track typology summary by algorithm.

Algorithm	Transient Laundering Path	Mixed / Uncertain	Stable Operator Cell
Louvain	58	12	0
Leiden	59	12	0
Infomap	62	26	0

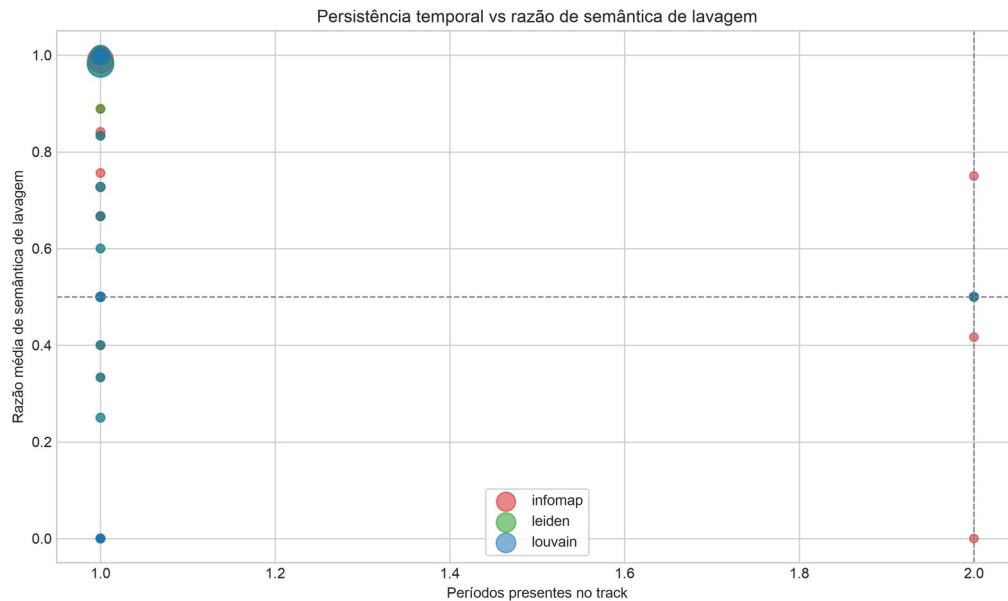
Source: results/paper_table_community_typology_summary.csv

Figure 22 – Community track typology distribution by algorithm.



Source: Authors (2026).

Figure 23 – Temporal persistence vs. laundering semantic ratio.



Source: Authors (2026).

3. CONCLUSION

Based on the reproducible pipeline applied to the dataset of addresses associated with GRU, FSB, and SVR, this study concludes that blockchain forensics delivers robust technical evidence for investigative prioritization, particularly when integrating structural graph metrics, operational temporality, and semantic transaction classification. The findings confirm strong flow concentration in a small subset of nodes, distinct high-intensity activity peaks, and a predominance of transient community paths with low quarter-over-quarter persistence.

All proposed research objectives were successfully fulfilled: the data baseline was enriched and validated, the directed graph was modeled with centralities, transaction volumes were evaluated across historical and current financial benchmarks, and analytical claims were strictly categorized into evidentiary certainty tiers to prevent over-attribution. Complementary analyses (ransomware/Conti, OP_RETURN, OSINT attribution, mixer pivoting) demonstrated high diagnostic utility for generating hypotheses while preserving evidentiary boundaries where no direct overlap was observed.

Regarding the research problem, operational inferences are feasible and valuable when bounded by methodological traceability and off-chain triangulation. Key limitations include inherent on-chain visibility constraints, absence of direct off-chain metadata, and the inability to equate addresses to civil identities without

corroborating evidence. Future work should expand cross-chain analysis, incorporate exchange KYC/sanctions intelligence, and implement formal probabilistic attribution scoring models.

3.1 Ethical, Legal, and Compliance Considerations

This study strictly avoids doxxing, publishes no personal identifiable information (PII), and does not assert direct legal culpability. It maintains an open, scientific focus on public ledgers and declassified documentation. Operational application of these methods must comply with applicable data protection laws, due process, international mutual assistance protocols, and digital chain-of-custody standards.

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